From History to Predictions: Machine Learning Approaches to Epilepsy Diagnosis

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Motivations and research question

Motivations

Epilepsy is a chronic neurological disorder characterized by unprovoked, recurrent seizures. Because the underlying causes are often unknown, diagnosis and treatment can be challenging and delayed. Early prediction of epilepsy risk would allow patients to receive treatment sooner, improving quality of life and reducing the risk of complications.

Problem statements

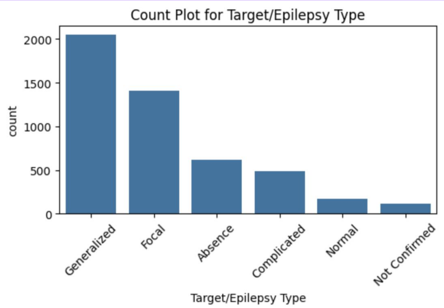
1) How can we better diagnose epilepsy?
2) What are the most influential symptoms in determining the presence and type of epilepsy?

Goal

Use clinical and seizure-related features to build, evaluate, and interpret machine learning models that predict epilepsy type and identify the symptoms that most strongly influence classification.

Data set

- Data set: Epilepsy Disorder Dataset (Kaggle)
- Total observations: 4856
- Total variables: 30
- 29 predictors, 1 response
- Response variable: Epilepsy diagnosis
- Categorical (5 possible outcomes): Absence, Normal, Complicated, Focal, Generalized



	Age	Gender	Weight	Height	Medication Status	Alcohol or Drug Use	EEG Abnormality Detected	MRI/CT Scan Result	Seizure Frequency	Seizure Duration	...	Flashing Lights Sensitivity	Loud Sound Sensitivity	Missed Medication	Family History of Epilepsy	Head Injury History	Brain Tumor
0	15	Male	48.3	168.2	On Medication	Yes	Yes	Normal	1	60	...	Yes	Yes	No	No	No	No
1	4	Female	56.0	174.5	On Medication	No	Yes	Abnormal	1	30	...	No	No	No	Yes	Yes	Yes
2	36	Female	44.8	156.2	On Medication	No	Yes	Abnormal	3	30	...	No	Yes	Yes	Yes	No	No
3	32	Female	70.4	180.4	Not on Medication	No	Yes	Not Done	0	60	...	Yes	Yes	No	No	No	No
4	29	Male	54.5	161.7	On Medication	No	Yes	Not Done	1	60	...	No	Yes	No	Yes	No	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
51	76	Female	58.1	170.3	On Medication	No	Yes	Normal	0	60	...	Yes	No	No	Yes	No	No
52	10	Male	80.6	172.9	On Medication	Yes	Yes	Abnormal	2	60	...	Yes	No	No	No	Yes	No
53	67	Male	37.0	159.6	On Medication	No	Yes	Normal	1	120	...	No	No	No	Yes	No	No
54	33	Male	54.7	156.5	On Medication	No	No	Normal	2	90	...	No	Yes	No	No	Yes	No

Data Processing

Type	Count	Description
Numeric	5	Quantitative measurements (age, weight, height, etc)
Categorical	4	Categorical variables with multiple levels
Binary	20	Categorical variables with 2 levels

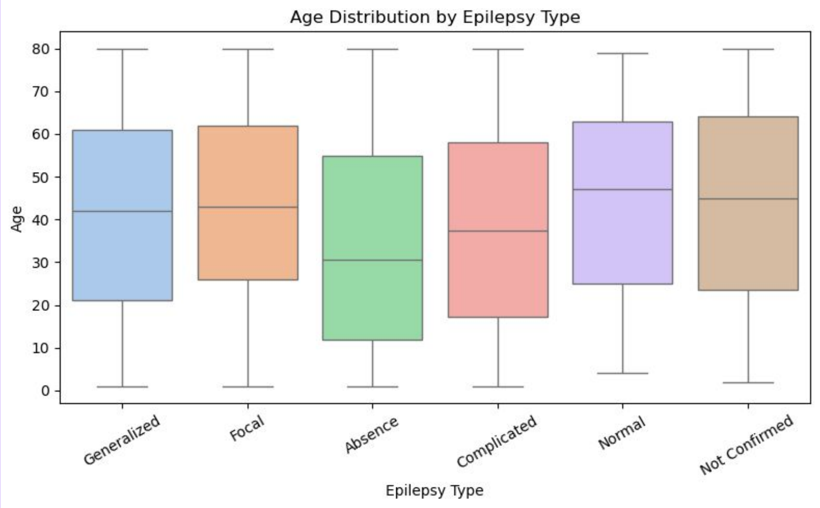
1) Numeric → Standardized (z-score scaling) to ensure fair weighting for models sensitive to Euclidean distance (e.g., logistic regression, KNN, SVM)

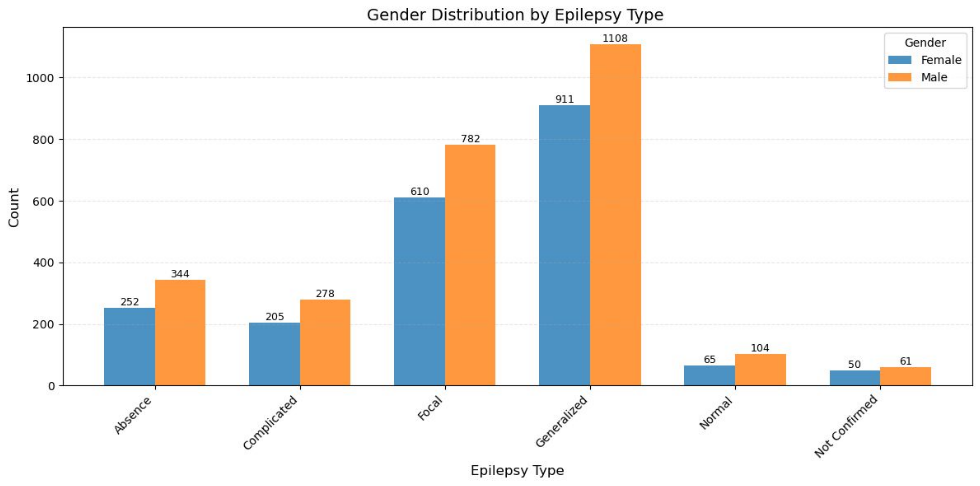
2) Categorical → One-hot encoded to convert multi-category variables into interpretable binary indicators

3) Binary → binary encoding with 1/0

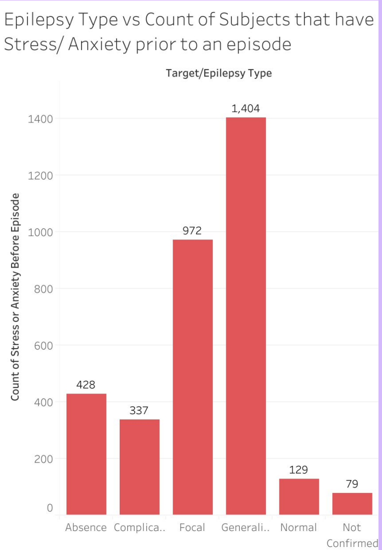
* All rows of data were retained as there are no missing values within the dataset

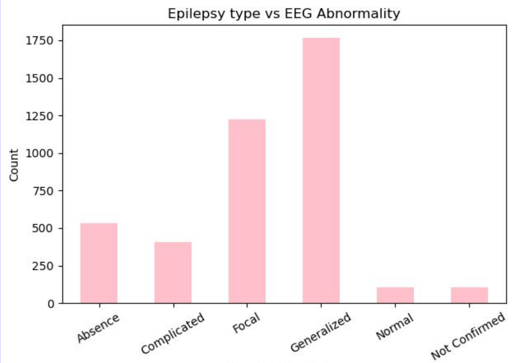
Demographic Distribution of the data

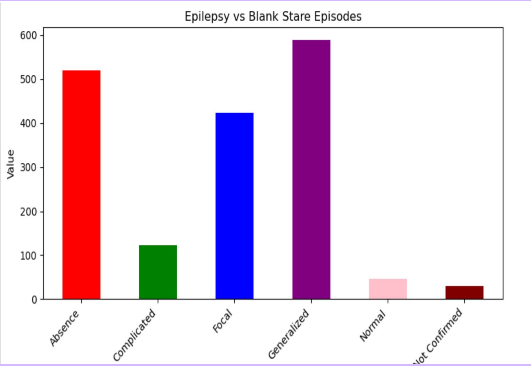


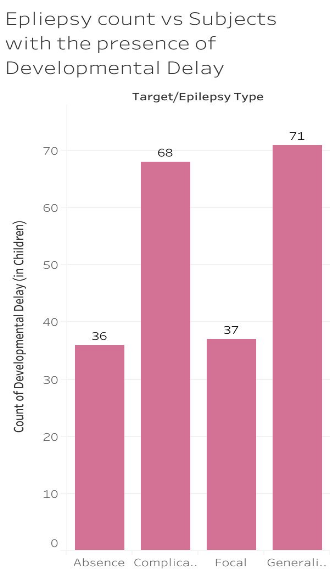


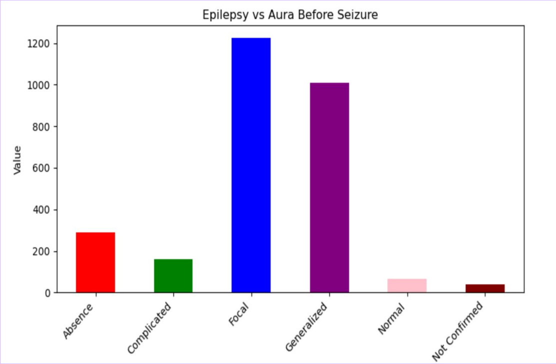
EDA - Distribution of Risk factors prior to Epileptic Episodes



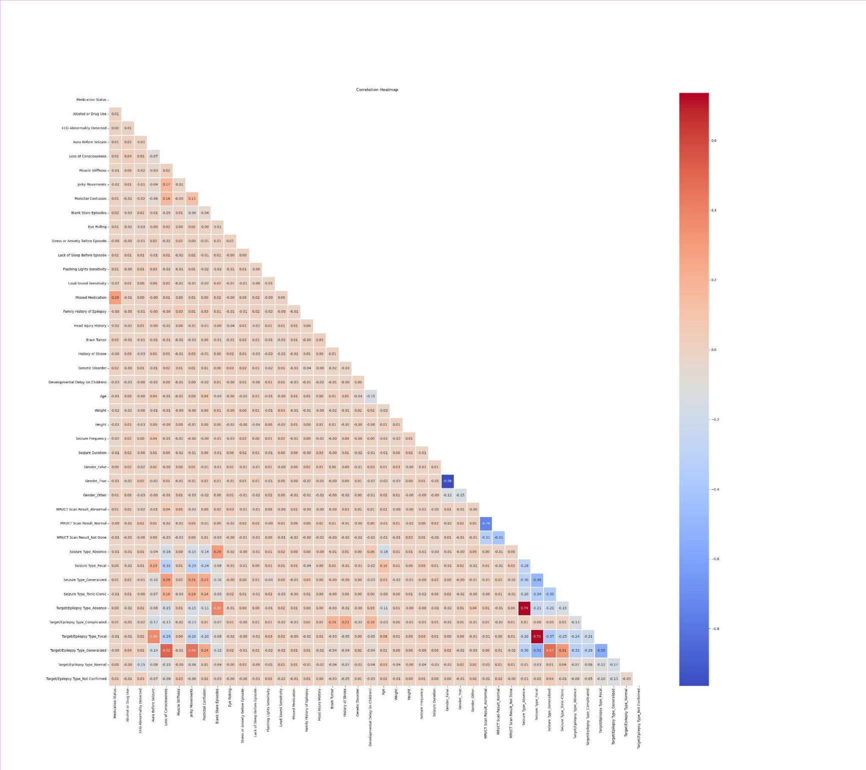




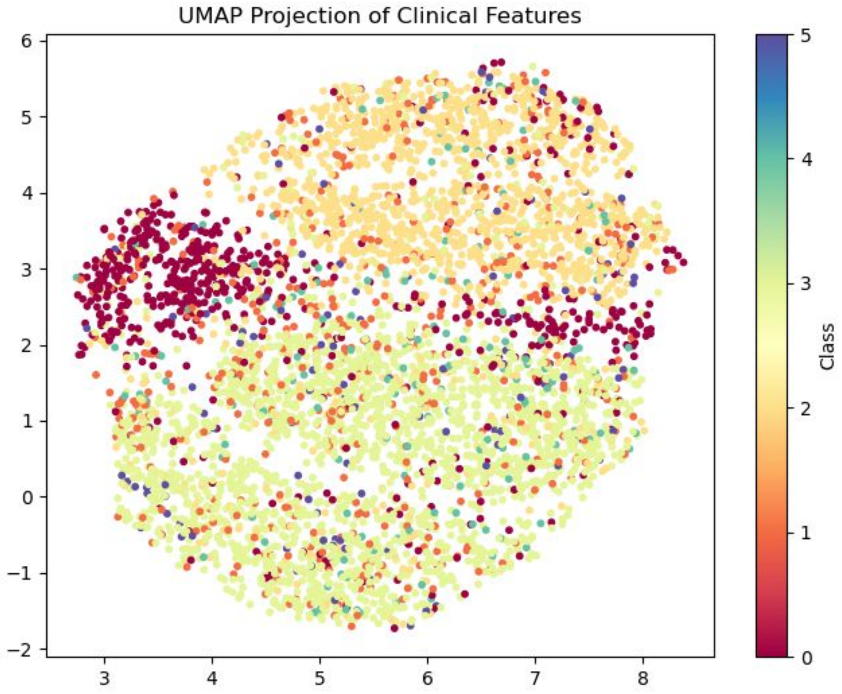




EDA - Initial understanding of the data



UMAP



Labels encoding:

0	Absence
1	Complicated
2	Focal
3	Generalized
4	Normal
5	Not confirmed

- Biggest class distinctions: Absence, Focal, Generalized
- Complicated, Normal and Not confirmed are scattered throughout

Overview of models tested

Model	F1 (weighted)	Accuracy
Gradient Boosting	0.8230	0.8143
Cat Boost	0.8117	0.8020
XGBoost	0.8128	0.8017
Logistic regression	0.7984	0.7751
Random Forest	0.7953	0.7689
SVM	0.7860	0.7534
k-NN	0.7233	0.6810

Decreasing accuracy

Summary of model results

Best model: Gradient Boosting

- Moderate data set - shallow tree from gradient boosting handles such datasets well
    - XGBoost and CatBoost tend to overfit
- Staged additive modeling with strong bias control helps to produce smooth decision boundaries often suitable for clinically structured features

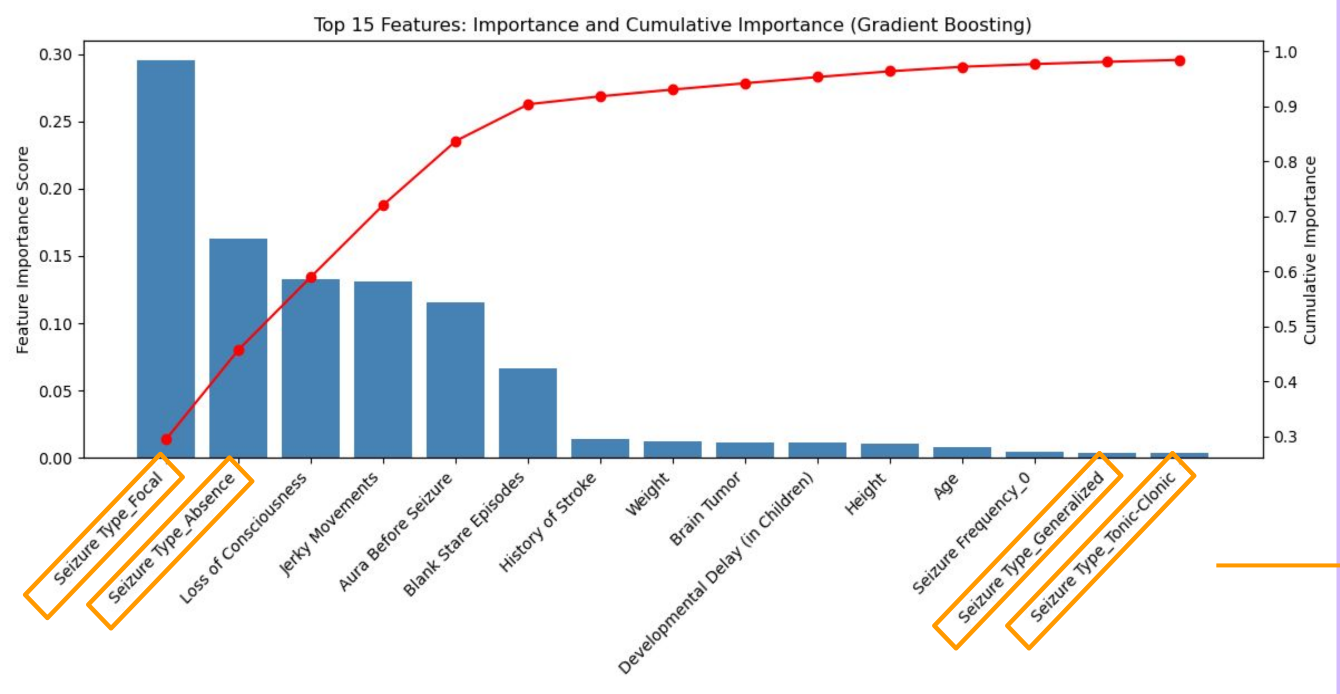
Worst model: k-NN

- Poor with binary features since it utilizes euclidean distances to determine similarity
- Sensitive to irrelevant features - does not weight features by importance
- Curse of dimensionality - relative distances collapse
- Poor decision boundaries for multi-class clinical data

Tuning of model hyperparameters

Model	Parameters	Best F1 (weighted)	Best accuracy	Best parameters
Gradient Boosting	n_estimators, learning_rate, max_depth	0.81489	0.82407	n = 200 lr = 0.05 depth = 3
XGBoost	n_estimators, learning_rate, max_depth, min_child_weights, subsample, reg_lambda, reg_alpha	0.8006	0.8189	n = 100 lr = 0.1 depth = 2 min_child_weights = 3 subsample = 0.8 reg_lambda = 5 reg_alpha = 0.5

Feature selection



Top **15** features explain **~98.3%** of the model's predictive signal

4/15 are one-hot encoded categories from the same variable, indicating that **seizure type** is crucial in determining the type & presence of epilepsy

Comparison of models with reduced data set

	Full model	Reduced model (top 15 features)	Difference between models
F1 (weighted)	0.81489	0.79372	-0.02117
Accuracy	0.824074	0.80761	-0.01646

- Top 15 features → only **~0.02** performance drop.
- Most predictive signal comes from a **small subset** of features (11 of the original variables)

Conclusion

The best model for predicting the presence of epilepsy: **Gradient Boosting**

- Weighted F1 = 0.815
- Accuracy = 0.823
- With hyperparameters: *n = 200, learning_rate = 0.05, depth = 3*

Out of **44 clinical features**, the **top 15 features** explained ~**98%** of the model's predictive signal

- Indicates that epilepsy type can be predicted from a small, clinically meaningful subset of symptoms
- **Seizure type** is the most important variable in predicting response

Key insight:
A compact set of clinically meaningful symptoms is sufficient for accurate epilepsy classification using Gradient Boosting

Thank you!

Do you have any questions?